

MYSQL DATABASE MANAGEMENT SYSTEM

LAB SHEET #6

Title:

- Implement group by, having clause and perform nested sub query in MySQL

Objectives:

- To understand and implement GROUP BY and HAVING clause
- To understand and use nested sub query

Background Concept:

DB Source: https://github.com/datacharmer/test_db

• GROUP BY

- It is used to divide records into groups.
- `SELECT <column_name> FROM <table_name> WHERE <condition> GROUP BY <column_name>;`

• HAVING

- It is used to apply restrictions on group. Like where clause is used to restrict single row, having clause is used to restrict group or collection of rows
- `SELECT <column_name> FROM <table_name> WHERE <condition> GROUP BY <column_name> HAVING <group_condition>;`

• SUB QUERY

- Sub query is a select statement which is nested with another select statement. First subquery is executed and returns result to outer query then outer query executes. Subquery can be placed at (i) Where clause, (ii) From clause, (iii) Having clause.
- Comparison operators are used in single row subquery (which returns single row or value)
- Special operators or multiple row comparison operators like IN, ANY, ALL are used in Multiple row subquery (which returns more than one row or values)
- `SELECT <column_name> FROM <table_name> WHERE <operator> (SELECT <column_name> FROM <table_name> WHERE <condition>);`

Lab works:

1. Count and display number of employees grouped by gender.
2. Show department name and count the number of employees in each department
3. Show department names with average salary per department
4. List job title and maximum salary for each job title
5. List department name and number of employees having more than 20,000 employees
6. List only those job titles and average salary having salary greater than 60,000/-
7. List department names and max salary of departments where maximum salary exceeds 1,00,000/-
8. List job title and minimum salary for each job title
9. Count number of employees hired each year
10. Count the number of employees per job title
11. List emp no, first name, last name of employees earning more than the average salary

12. List first name, last name of employees earning the maximum salary (without using join)
13. Show department names having employees with salary above 90,000/-
14. List first name, last name of the employees who are the department managers
15. List first name, last name of employees who never became managers

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